



ISO 14001:2015

ISO 9001:2015

ELECTRO MECH[®]

JOURNEY TO LEADERSHIP:
YOUR ONE STOP DESTINATION
FOR EVERY POWER SOLUTIONS



COMPANY PROFILE

ELECTROMECH
AUTOMATION & ENGINEERING LTD.

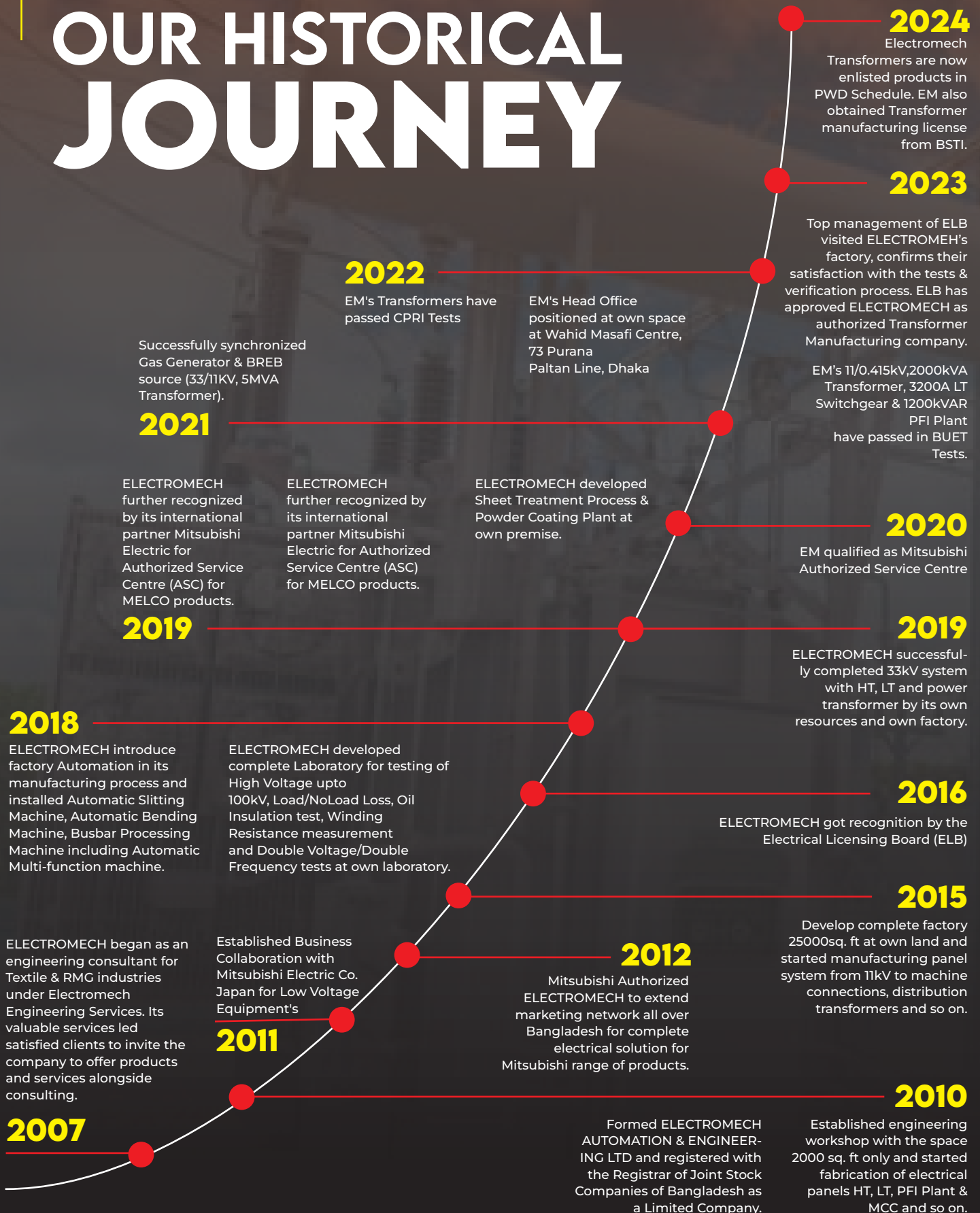
ELECTROMECH POWER LTD.



www.electromechbd.net



OUR HISTORICAL JOURNEY



ABOUT US

We create a great working environment where we all try to cultivate a strong culture that will help us to meet the challenges of the present & future. We are passionate about our business, customer & people. We strive to create a real sense of partnership with our clients; we try to develop human resources to reach their potential. We together always think for the green environment, a greener world.

MISSION:

To provide a level of service that is beyond the expectations of our customer's, maintain highest level of quality. Maintain highest degree of professionalism in offering products.

VISION:

We create a great working environment where we all try to cultivate a strong culture that will help us to meet the challenges of the present & future. We are passionate about our business, customer & people. We strive to create a real sense of partnership with our clients; we try to develop human resources to reach their potential. We together always think for the green environment, a greener world.

VALUES:

At ELECTROMECH, we uphold integrity through ethical standards and open communication, fostering lasting relationships. We prioritize efficiency by delivering cost-effective, high-quality solutions. Teamwork drives our productivity, with mutual respect and diverse expertise shaping every project. Professionalism defines us, emphasizing competence, dependability, and excellence. Our commitment to exceptional customer service ensures we exceed client expectations, setting us apart in the industry.

ORGANIZATIONAL STRUCTURE



PRODUCTS SUMMARY

MV Switchgear Panel

- 12KV Metal Clad & Metal Enclosed Panels
- Rated current upto 2000A
- High Breaking Capacity



MV Switchgear Panel

Power Transformers

- 33/11KV Voltage Class Manufacturing starting from 4MVA onwards
- OLTC
- Enhanced protection features



Power Transformers

Distribution Transformers

- 11KV/415V/220V Voltage Classes
- 100KVA to 4000KVA Capacity
- NLTC / OLTC Options
- Indoor / Outdoor Installation (suitable for Utility, Industry & Infrastructure)
- Oil / Dry Cast Resin Type
- Transformer



Distribution Transformers

LV Switchgear Panel

- Rated Insulation voltage 1000V
- Rated Current upto 10000A
- Panel Separation form upto 4B type 2
- Highly Customizable



LV Switchgear Panel

Automatic Power Factor Correction (APFC / PFI) Panel

- Rated capacity 60KVAR to 2400KVAR
- Maintains high Power Factor consistency
- Fuse or C.B. Protection option
- 6, 12 & 14 Stage Correction Options
- Harmonic Management by Detuned Reactors

Automatic Power Factor Correction (APFC/PFI) Panel



Diesel Gensets

- 50KVA to 800KVA Capacity
- Soundproof Body & Room Canopy
- Synchronizing Relay for parallel operation



Diesel Gensets



Factory Automation & Building Management System (BMS)

Factory Automation & BMS

- VFD (Inverters)
- HMI, PLC, Servo, SCADA
- Sensors, Controller, Timer, Counter, Meters
- Protection Relays

MV & LV Components



MV & LV Components

- 33KV & 11KV Vacuum Circuit Breakers
- ACB, MCCB, MCB & RCCB
- Magnetic Contactors
- Power Capacitors

Motor Control Centre (MCC)



Motor Control Centre (MCC)

- Fully customizable with automation
- Motor Starting, Controlling, Protection features

Busbar Trunking System



Busbar Trunking System

- 25A to 6300A
- Copper & Aluminium Conductor
- Steel & Aluminium Housing
- Upto IP67 protection for submersed operation

PRODUCTS DETAILS



MV Switchgear

Application: High Tension (HT) Power Distribution in Industries, Utilities, and Factories for safe and efficient control, protection, and isolation of electrical equipment. Suitable for both indoor and outdoor installations.

Standards: IEC 62271-100 & IEC 62271-200

Parameters:

- Rated Voltage 11KV & 33KV
- Rated Capacity: 630A to 2000A

Features:

- Metal Clad & Metal Enclosed Panels for enhanced safety and reliability
- Protection Options: Instantaneous & IDMT Overcurrent protection, Over & Undervoltage, Earth
- Fault Protection, Differential Protection (Optional)
- Trip Circuit supervision Relay with Master Trip relay for real-time monitoring and remote operation
- Low system losses and reduced maintenance requirements



Power Transformers

Application: Medium Voltage Power Distribution in Industries / Utility / Factories. Used mainly for Step Down operation. Designed for use in both Outdoors and indoors.

Standards: BDS IEC 60076-1: 2016

Parameters:

- Rated Voltage: 33KV, 11KV
- Rated Capacity: 3000KVA to 12500KVA
- Vector Group: Dyn11 or as per demand

Features:

- On Load Tap Changer (OLTC) / No Load Tap Changer (NLTC) options
- ONAN / ONAF application
- Enhanced Protection with Buchholz Relay & Differential Relay (Optional)
- Low Impedance
- Lower No Load & Full Load Loss
- Lower Greenhouse Emissions
- Increased Life of Operation
- Compliant to Standards & Regulatory requirements



Distribution Transformers

Application: Medium & Low Voltage Power Distribution in Industries / Utility / Factories/ Residential & Commercial Buildings. May be used mainly for Step Up / Down operation. Designed for use in both Outdoors and indoors.

Standards: BDS IEC 60076-1: 2016

Parameters:

- Rated Voltage: 33KV, 11KV
- Rated Capacity: 100KVA to 4000KVA
- Vector Group: Dyn11 or as per demand

Features:

- On Load Tap Changer (OLTC) / No Load Tap Changer (NLTC) options
- Oil Immersed Type Transformers
- Dry Type Cast Resin Type Transformers also available on request
- All routine test facility at Electromech Manufacturing plant
- Low Impedance. Lower No Load & Full Load Loss
- CPRI- India & BUET Type tested
- Increased Life of Operation
- Possible to Synchronize with Generators
- Compliant to Standards & Regulatory requirements





LV Switchgear Panel

Application: Safe and efficient power distribution, control, and protection in industrial, commercial, and utility settings. Suitable for both indoor and outdoor installations, ensuring reliable operation of electrical systems.

Standards: IEC 61439-1, IEC 60947-2

Parameters:

- Rated Voltage: Up to 690V
- Rated Capacity: Up to 10000A

Features:

- Equipped with high-quality circuit breakers, contactors, and relays sourced from leading manufacturers for superior performance
- Configurable as Main Distribution Panels (MDP), Sub-Distribution Panels (SDP), and Motor Control Centres (MCC)
- Overload, short-circuit, and earth fault protection.
- Robust, modular, and compartmentalized design with Form 2a/2b, 3a/3b, or 4a/4b separation options for increased safety and reliability
- Mechanical interlocks, and insulation barriers for maximum safety
- Tailored configurations available to meet specific project requirements, including Bus Coupler Panels

Motor Control Centre (MCC)

Application: Centralized control of electric motors in industries, utilities, and factories. Used for efficient motor operation, protection, and monitoring in various applications such as pumps, conveyors, and fans suitable for process automation.

Standards: IEC 61439-1 & IEC 61439-2

Parameters:

- Rated Voltage: 415V
- Rated Capacity: Up to 6300A

Features:

- Unlimited possibility with fully Automated panel with integrated VFD, PLC, HMI, and SCADA systems
- Robust, modular, and compartmentalized design with Form 2, 3, or 4 separation options
- Soft starters and VFD integration for reduced power consumption and smooth motor start/stop
- Real-time monitoring of motor parameters like current, voltage, and power factor
- Configurable as per client-specific requirements, including different starter types (DOL, Star-Delta, Soft Starter)



MV & LV Components



Application: High-performance Medium Voltage (MV) and Low Voltage (LV) components for reliable power distribution, control, and protection. Advanced technology for high reliability and consistent performance.

Standards: IEC 60947, IEC 62271

Products:

- MV Components: 11 & 33KV Vacuum Circuit Breakers (VCB), Load Break Switches, Protection Relays
- LV Components: Air Circuit Breakers (ACB), Molded Case Circuit Breakers (MCCB), Miniature Circuit Breakers (MCB), Contactors, Overload Relays
- Rated Capacity: All Standard Ratings

Features:

- Comprehensive protection against overcurrent, short-circuit, earth fault, and phase loss
- Low power losses with energy-saving features, reducing overall operational costs
- Tailored solutions available to meet industry-specific needs, including options for factory automation and smart power systems



Automatic Power Factor Correction Panel (APFC)

Application: Designed to improve power quality by automatically managing and correcting the power factor in industrial, commercial, and utility installations. Ideal for reducing energy consumption, minimizing reactive power penalties, and enhancing the efficiency of electrical systems.

Standards: IEC 60831, IEC 61439

Parameters:

- Rated Voltage: Up to 690V
- Rated Capacity: 50 kVAR to 2400 kVAR
- Control Stages: Upto 16 fully customizable stages

Features:

- Equipped with Heavy Duty, low-loss capacitors that provide stable and reliable performance with self-healing properties
- Intelligent power factor controller with real-time monitoring and automatic step switching for optimal correction
- Reduces reactive power demand, leading to lower energy costs and improved power factor
- Harmonic filtering options to reduce distortion and prevent equipment damage
- Built-in protection systems including fuses, circuit breakers, and disconnect switches for safe operation
- Optimizes power factor to near unity, reducing demand charges and improving overall energy efficiency



Busbar Trunking System (BBT)



HP-K: HIGH POWER DISTRIBUTION

- 400A-6300A Range
- 4P – TPN+PE (Housing) / 5P-TPN + PE (Bar)
- IP55 (Indoor) / IP67 (Outdoor)
- Suitable at high Ambient Temp- 40°C



MP-H: MEDIUM POWER DISTRIBUTION

- 160A-600A Range
- 4P – TPN+PE (Housing) / 5P-TPN + PE (Bar)
- IP55 (Indoor)

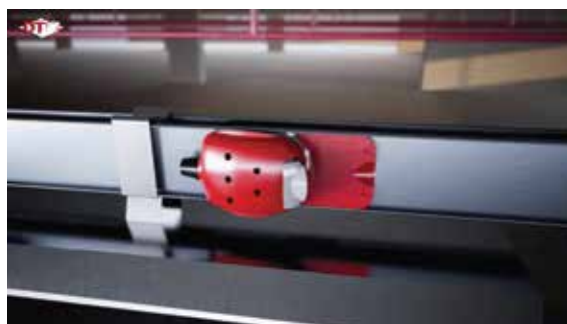


SP-H: SMALL POWER DISTRIBUTION

- 100A-160A Range
- 4P – TPN+PE (Housing) / 5P-TPN + PE (Bar)
- IP55 (Indoor)

AB-E: LIGHTING & MACHINE BBT

- 25A-32A-40A-63A Range
- 4P – TPN+PE (Housing) / 5P-TPN + PE (Bar)
 - IP55 (Indoor)
- Supports Both Plugs (Direct or Fused) & Tap Off Boxes



Factory Automation & Building Management System (BMS)



Application: Comprehensive solutions for enhancing operational efficiency in industrial automation and intelligent building management. Ideal for factories, commercial buildings, and smart facilities requiring precise control, monitoring, and optimization of processes.

Standards: IEC 61131, ISO 16484

Features:

- Factory Automation Solutions with Programmable Logic Controllers (PLC), Human Machine Interface (HMI), Variable Frequency Drives (VFD) & SCADA Systems
- Building Management System (BMS) for HVAC Control, Lighting Control, Access Control, Energy Monitoring and Fire & Safety Systems
- Connectivity support for various Communication Protocols such as Modbus, BACnet, Ethernet/IP for seamless integration with existing systems
- IoT & Cloud Integration enables remote monitoring and control capabilities via IoT platforms for enhanced flexibility
- Advanced data analytics for predictive maintenance and process optimization
- Tailored solutions to meet specific requirements of industrial and commercial applications, ensuring optimal performance and efficiency
- Significant energy savings and reduced operating cost

Diesel Gensets

Application: Reliable power backup solutions for industrial, commercial, and residential applications. Ideal for ensuring uninterrupted power supply during outages and peak demand periods. Suitable for use in both indoor and outdoor environments.

Parameters:

- Rated Capacity: 50 kVA to 800 kVA
- Design: Soundproof Body & Room Canopy for noise reduction

Features:

- Integrated soundproof body and room canopy, reducing noise levels to comply with local regulations and enhance user comfort
- Built-in safety features such as automatic voltage regulation (AVR), overload protection, and emergency stop functions
- Optional ATS for automatic switching between mains and generator power, ensuring a smooth transition during outages
- Synchronizing relay enables seamless parallel operation, load sharing and enhanced power availability
- Optimized fuel consumption for cost-effective and long-lasting performance
- User-friendly digital control panel for easy monitoring and operation, including key parameters like voltage, current, frequency, and running hours
- Durable and weather-resistant design for optimal performance in harsh environments
- Available spares for easy troubleshooting and maintenance



EPC (Engineering, Procurement and Construction)



ELECTROMECH is a promising Engineering, Procurement, and Construction (EPC) company in Bangladesh, specializing in delivering high-quality, fast-paced power and energy support solutions. Backed by a wealth of in-depth knowledge and expertise, our dedicated team ensures reliable, compliant, and efficient service in every aspect of our operations. Our processes are rigorously audited to maintain the highest standards of company conformity. With extensive experience in implementing projects of varying scales, we collaborate with multinational companies as a consortium, joint venture partner, or local agent, consistently delivering exceptional service and solutions.

Some of our Govt./EPC Contracts:

- **PDB:**
 - 11/0.4 kV, 250 kVA Distribution Transformer- 50 Nos. Hundred Percent Reliable and Sustainable Electrification of Hatiya Island, Nijhum Island & Kutubdia Island
 - Procurement of Oil Deflector (4 set shaft seal assembly) of Gas Turbine Generator of Chandpur 150 MW CCPP
 - Supply of various capacities TP MCCB for emergency requirements for the various residential quarter & Bidyut Bhaban Under Civil Construction Division-2, BPDB, Dhaka.
- **EED:**
 - Supply & Installation of 2 X 2000 kVA Substation of "Development of Bangladesh University of Professionals (Phase 1) (1st Revised)"
 - Supply and Installation of 2500 kVA Substation with ancillary civil works in West Campus of BUP under the Project of "Development of Bangladesh University of Professionals (2nd Revised)"
 - Supply and Installation of 1250 kVA Substation with HT and LT Cable for Utility Building under the Project of "Development of Bangladesh University of Professionals (2nd Revised)"
- **DESCO:** Supply of Transformer Protection Equipment & Capacitor Bank for Dhaka Power System Expansion and Strengthening Project in DESCO Area
- **NESCO:** 33KV Vacuum Circuit Breaker (VCB), Procurement of 33 kV Current Transformer (CT)

FACTORY MODERNIZATION & PROCESS IMPROVEMENT



Introduction of Treatment Plant & Powder Coating Painting Process is the new era in panel system of ELECTROMECH which complies 8 processes instead of 7 processes.



Multi-function Machines, Brings Precision Into The Electromech Manufacturing Process



It's Difficult & Often Impossible To Remain & Retain The Physiognomies Of Copper Busbar After It's Bending As Per Functional Requirements. The Busbar Process Machines Has Relaxed The Busbar Bending Team By The Introduction Of This Programmable Busbar Processing Machine.



Introduction Of Automatic Cnc Cutting & Bending Machines Has Been Differentiated The Em Panel Systems Fabrication Jobs Than Others In The Market.

MAJOR COMPLETED PROJECTS



EM LV Panel &
BBT System Installed for
Akij Textiles Mills Ltd.



2 x 33/11 kV 5 MVA
Complete Solution including
HT, LT Switchgear and BBT in
**Malek Spinning Mills Ltd. &
Knit Asia Ltd. (New Asia Group)**



33/11 kV 7.5 MVA &
2 nos 11/0.415kV 4 MVA
Substation Solution for
Euro Knit Spinning Ltd



11/0.415kV, 2000kVA
Substation Including 11kV
Automatic Voltage Regulator,
2000kVA Transformer &
11kV Bypass Switch at
Knit Concern Group



Complete 33/11KV
Substation & Gas Gen.
Sources Have Synchronised For
JM Fabrics (New Asia Group)





33/11 kV
Complete
Substation in
United group



LV Switchgear Panels
with all equipments from
Mitsubishi range of
components for
Walton



EM LAB

Quality is a never ending challenge. It is a continual improvement process. No scope of ambiguity and or vagueness in the quality process. Thus, quality shall be conforming to standard, measurable and tangible in nature.

To test verify & validate the designed parameters quoted to customer are tested in ELECTROMECH's own laboratory having capacity to measure & reports on:

1. No Load Loss Test
2. Load Loss Test
3. Turns Ratio Test
4. Winding Resistance Test
5. Double Voltage Test
6. Double Frequency Test
7. High Voltage Test upto 100kV



Chief Inspector of ELB with his team visited ELECTROMECH Factory. ELB has approved & Issued Transformer manufacturing License to EM at very first visit.

CERTIFICATIONS



ISO 9001_2015



ISO 14001_2015

PWD Enlistment for
Substation WorksBSTI Certificate
for TransformerCPRI Type Test -
200KVA TRFCPRI Type Test -
250KVA TRFBUET Test - 2000
KVA TransformerBUET Test -
3200A LT PANELBUET Test -
1200KVAR PFIBUET Test -
11KV HT PANEL

MIST Test Certificate

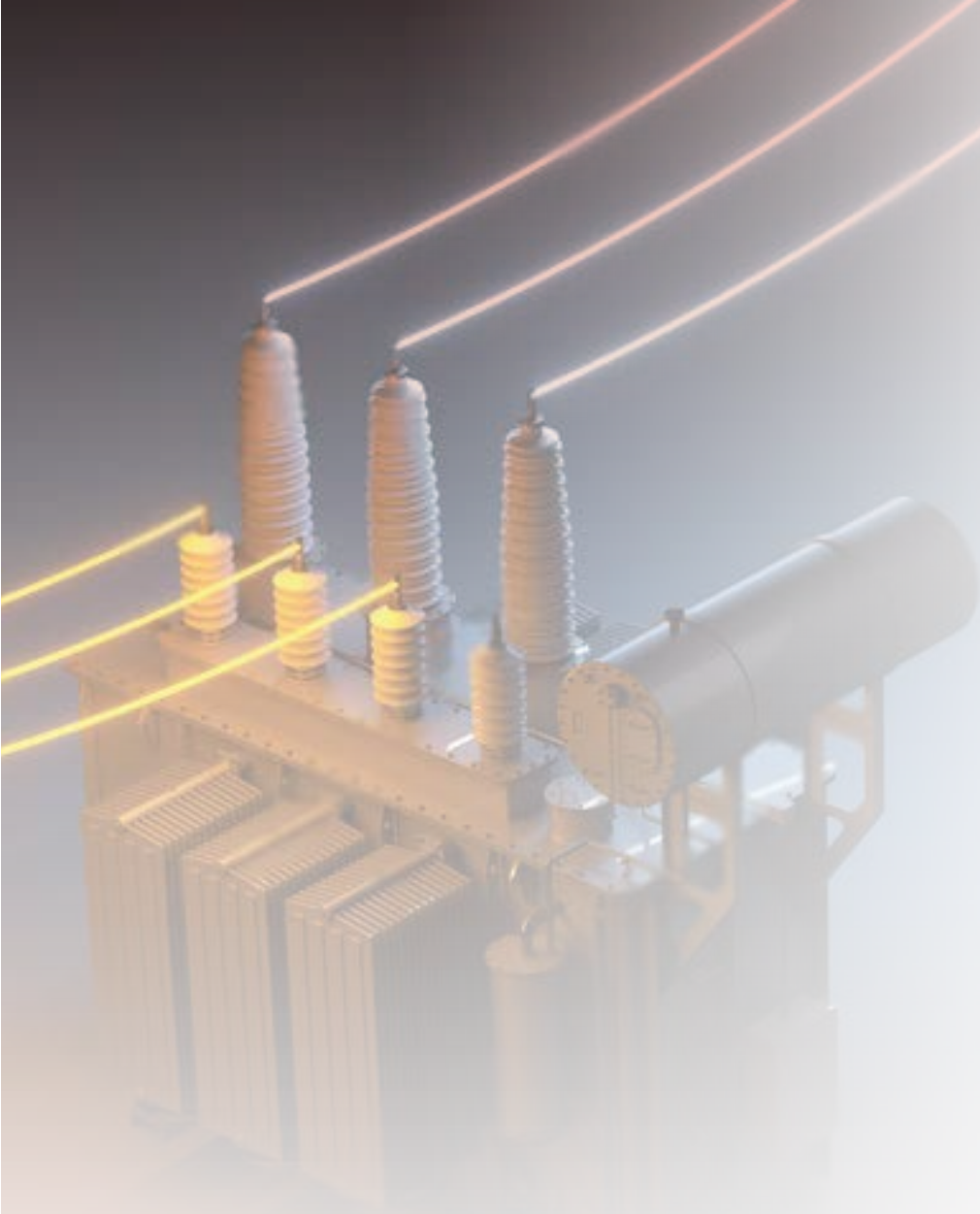
Approval for Electrical
Substation Equipment
Manufacturer

OUR CLIENTS



MAH SPINNING





ELECTRO MECH®



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